



Advanced Log Analysis Report

Overview

This project demonstrates core SOC analyst capabilities including log correlation, anomaly detection, and log enrichment using Elastic Security, Security Onion, and Google Sheets.

Log Correlation

Objective

Correlate failed login attempts (**Event ID 4625**) with outbound network traffic to identify suspicious activity.

Method

- Ingested sample logs into Elastic
- Queried:

event.code:4625 AND destination.ip:*

Results

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS request
2025-08-18 12:01:00	4625	192.168.1.100	45.33.32.156	Possible C2 communication
2025-08-18 12:02:00	4625	192.168.1.101	1.1.1.1	Repeated failed attempts

Analysis

Multiple failed login attempts followed by outbound connections from the same source IP indicate potential brute-force activity followed by external communication, suggesting possible compromise.

2. Anomaly Detection

Objective

Detect high-volume outbound traffic indicating potential data exfiltration.

Rule Configuration

- Rule Type: Custom Query
- Query:

network.bytes_out > 1000000



- Time Window: 1 minute
- Index: bots-practice

Testing

Simulated high-volume traffic using mock data exceeding 1MB.

Result

The rule successfully triggered alerts for events exceeding the threshold.

Analysis

Unusual spikes in outbound traffic may indicate data exfiltration or unauthorized transfers.

3. Log Enrichment (GeoIP)

Objective

Enhance logs with geographic context for better threat visibility.

Method

- Created ingest pipeline with GeoIP processor
- Enriched source.ip field with location data

Outcome

Added:

- Country
- City
- Geolocation coordinates

50-Word Summary

GeoIP enrichment mapped IP addresses to geographic locations, adding country and city information to logs. Analysis revealed outbound connections to unfamiliar regions following repeated failed login attempts. This additional context improved threat visibility, enabling faster detection of suspicious behavior, potential command-and-control activity, and possible data exfiltration from compromised systems.

Conclusion

This project successfully demonstrated how log correlation, anomaly detection, and enrichment techniques can identify suspicious activity and improve visibility in a SOC environment. These methods are essential for detecting potential compromises, unauthorized access, and data exfiltration attempts.

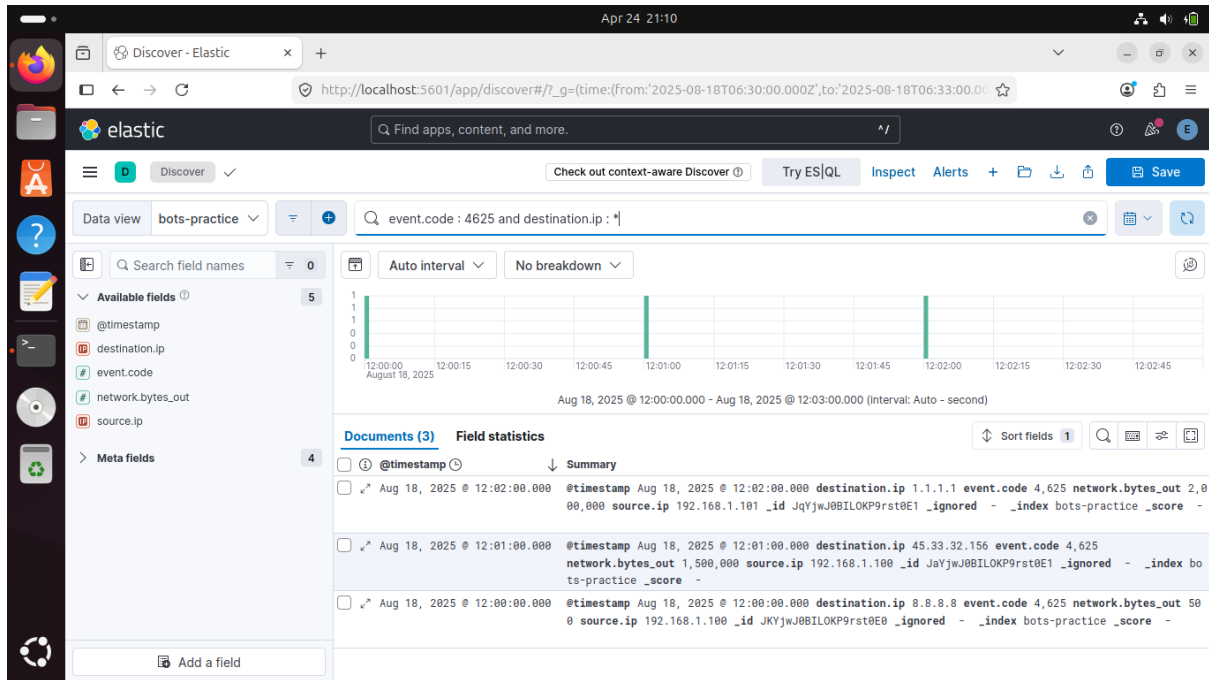


Fig 1: Log correlation

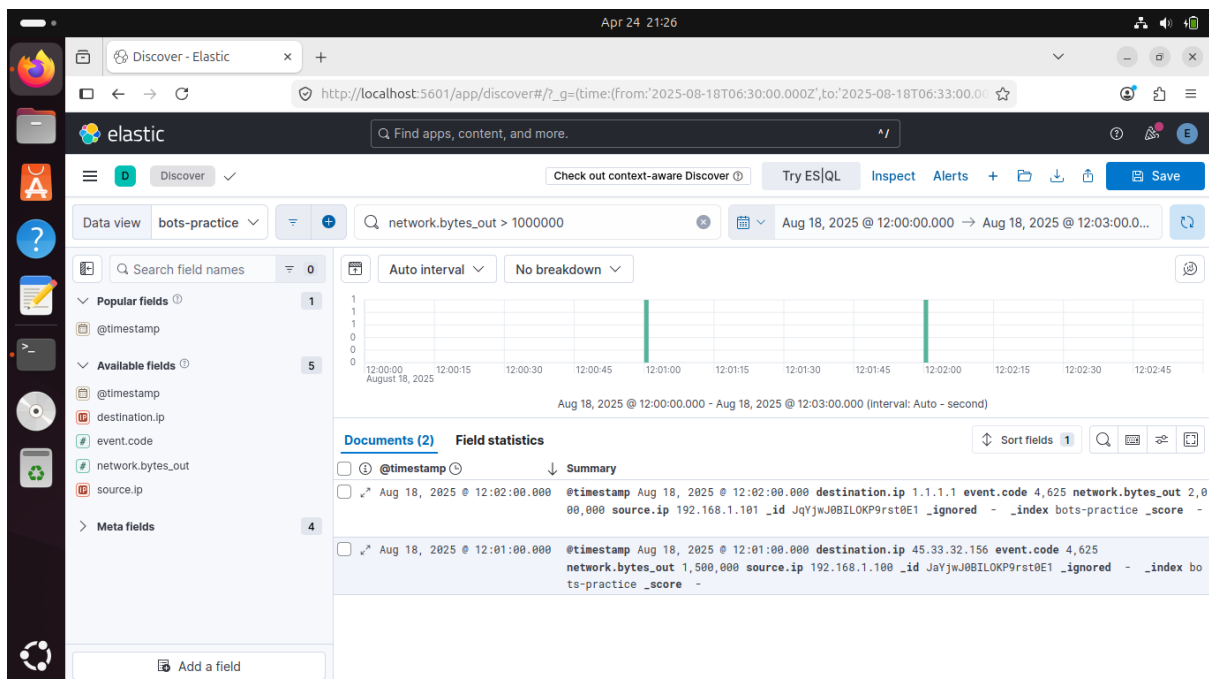


Fig 2: Anomaly Detection



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